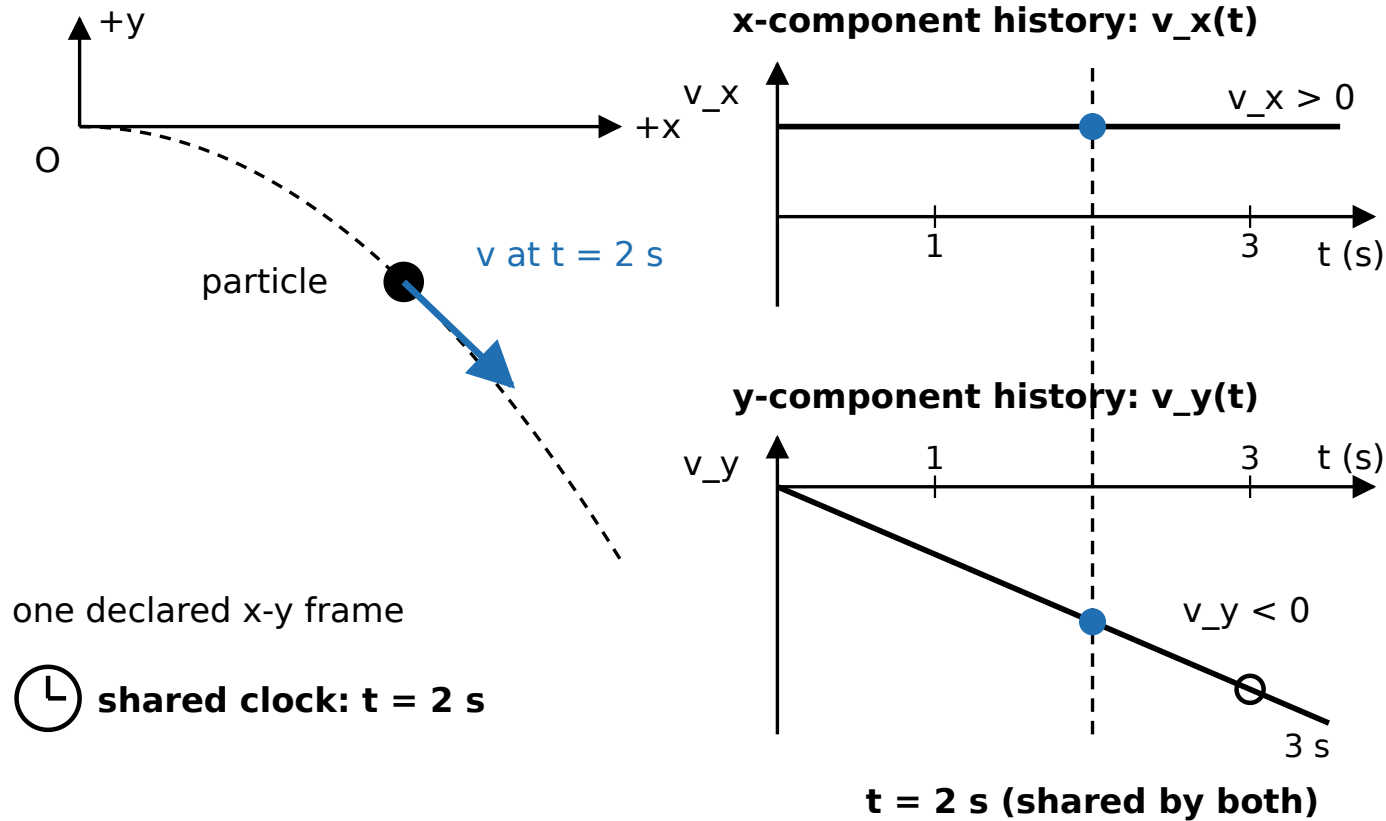


**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically:
Orientation map**

Split one plane-motion event into independent x/y components with one clock

Hard transition

Perpendicular components may be solved separately because each axis equation contains only that axis's quantities, but they still describe one object at one physical time.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Governing relation

$$\Delta x = u_x t + 0.5 a_x t^2; \Delta y = u_y t + 0.5 a_y t^2$$

Each perpendicular displacement component follows the 1D constant-acceleration displacement relation over one common interval.

- a_x and a_y are constant over the interval.
- Both component equations use the same elapsed time and frame.

Compact anchor

A particle leaves the origin with $u = (2, 0)$ m/s and constant $a = (0, 6)$ m/s². Where is it at $t = 1$ s?

$\Delta x = 2 \times 1 = 2$ m and $\Delta y = 0.5 \times 6 \times 1^2 = 3$ m, both read at the same $t = 1$ s, so it is at (2, 3) m.

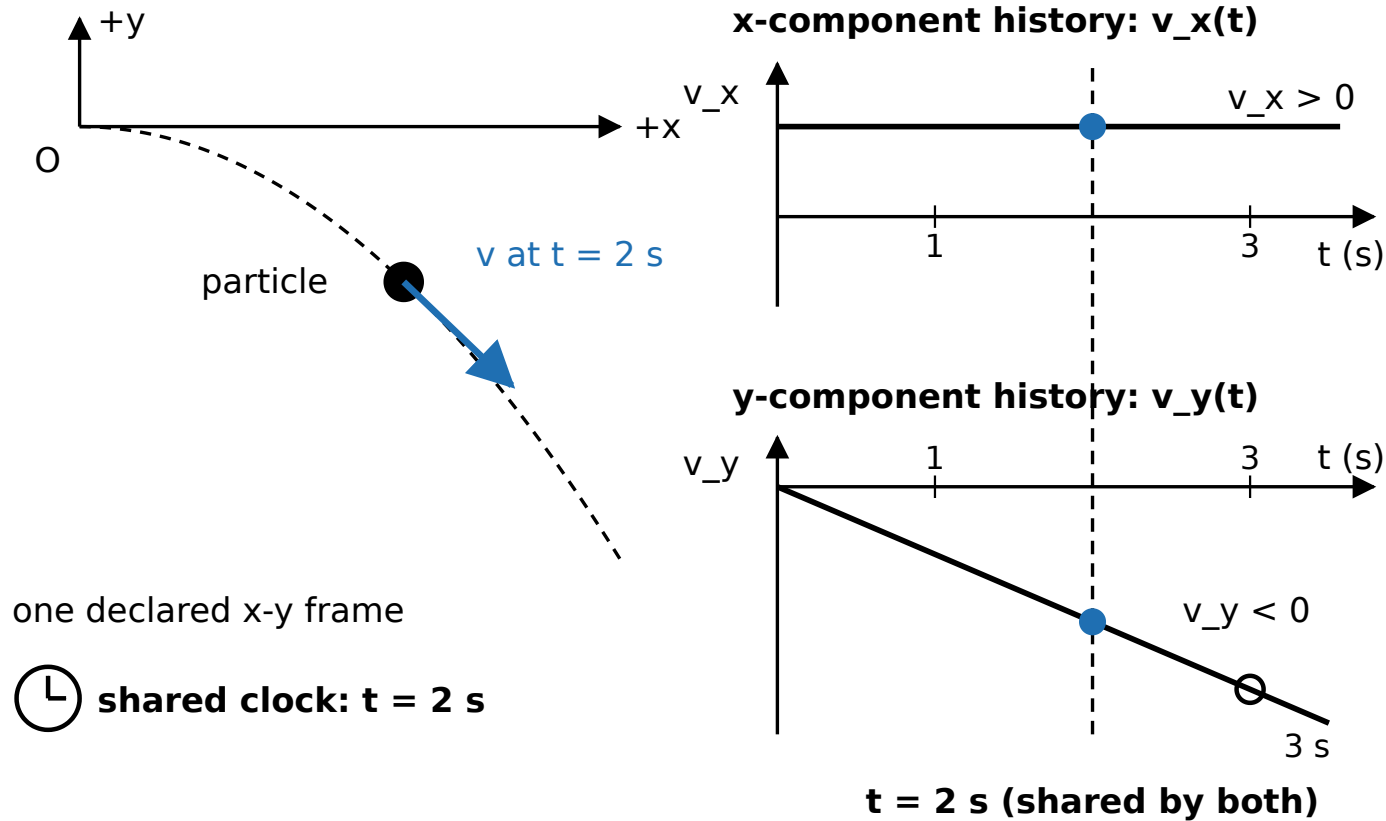
Before you go on

A particle has x-velocity 6 m/s and y-velocity -8 m/s at $t=3$ s. Explain what 'independent components' means and what must remain shared between the x and y descriptions.

Apply constant-acceleration kinematics component by component

Hard transition

The one-dimensional constant-acceleration equations can be applied to each perpendicular component independently when a_x and a_y are each constant over the same interval.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Governing relation

$$v_x = u_x + a_x t; \quad v_y = u_y + a_y t$$

Each perpendicular velocity component follows the 1D constant-acceleration relation over one common interval.

- a_x and a_y are constant over the interval.
- Both component equations use the same elapsed time and frame.

$$\Delta x = u_x t + 0.5 a_x t^2; \quad \Delta y = u_y t + 0.5 a_y t^2$$

Each perpendicular displacement component follows the 1D constant-acceleration displacement relation over one common interval.

- a_x and a_y are constant over the interval.
- Both component equations use the same elapsed time and frame.

Compact anchor

$u = (1, 2) \text{ m/s}$ and $a = (2, -2) \text{ m/s}^2$, both constant. What is the velocity at $t = 2 \text{ s}$?

$v_x = 1 + 2 \times 2 = 5$ m/s and $v_y = 2 - 2 \times 2 = -2$ m/s on the same clock, so $v = (5, -2)$ m/s.

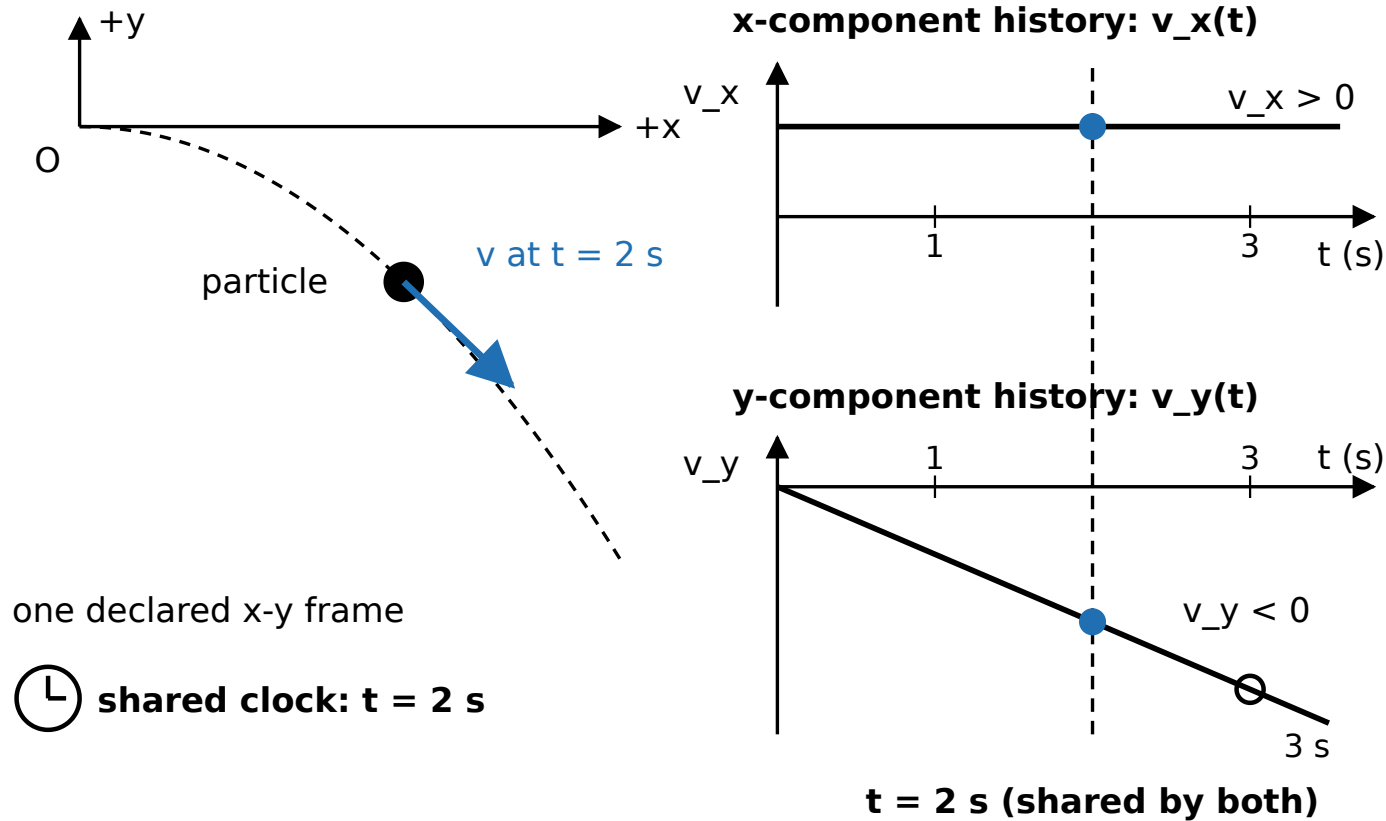
Before you go on

A particle starts with $u=(0,5)$ m/s and constant $a=(10,4)$ m/s². Set up the x and y position equations at common time t without solving a particular numerical t.

Select the gravity-only projectile model before using projectile consequences

Hard transition

After release in the ideal near-Earth model, projectile motion is not a new set of laws: it is the same component kinematics with $a_x=0$ and $a_y=-g$ (for +y upward).



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Governing relation

$$a_x = 0; a_y = -g; v_x = u_x; v_y = u_y - g t; \Delta x = u_x t; \Delta y = u_y t - 0.5 g t^2$$

With +y upward, air resistance neglected and near-Earth constant g, projectile motion is 2D constant acceleration specialized to zero horizontal and downward vertical acceleration.

- Near-Earth region with approximately constant g.
- Air resistance neglected.
- +y is declared upward for the displayed signs.
- The object is in free flight after release.

Compact anchor

An ideal projectile has $u_x = 10$ m/s and $u_y = 5$ m/s, with +y up and $g = 10$ m/s². Where is it, and how is it moving, 1 s after release?

$\Delta x = 10$ m and $\Delta y = 5 \times 1 - 5 \times 1^2 = 0$ m, so it is back at launch height; $v = (10, 5 - 10) = (10, -5)$ m/s.

Before you go on

A stone is launched horizontally from a cliff with speed u . With $+y$ upward and air resistance neglected, write its component acceleration, velocity and displacement relations after release and explain why one common t is used.

Physics · PRODUCT-PHY-KIN-2D-MOTION